EVERY FINITE MODULE OVER A PRINCIPAL IDEAL DOMAIN IS A FINITE FREE MODULE PLUS A SUM OF QUOTIENTS BY NONZERO ELEMENTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author writes '(small detail omitted)' twice inside the counted proof span and writes 'Since R/fR is Noetherian of dimension 0 (small detail omitted) we see that $R/fR = \prod R_j$ is a finite product of Artinian local rings R_i', using j and i inconsistently; both omissions and the inconsistent index are reproduced unchanged. The four references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- The splitting of a finite module over a Bezout domain into a free part and a torsion part. invoked at proof lines 37897-37898. Reduces the statement to the torsion case.
- A zero-dimensional Noetherian ring is a finite product of Artinian local rings. invoked at proof lines 37901-37904. Produces the factors R_j over which the module decomposes.
- The structure of finite modules over a generalized valuation ring. invoked at proof lines 37910-37911. Gives each M_j as a direct sum of cyclic quotients over R_j.
- Lifting cyclic quotients: choosing g_{ji} with (g_{ji}) = (f_j, f_{ji}). used at proof lines 37912-37913. Is the step that turns the decomposition over R/fR into one over R.

Source reference. Original lemma 00KJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 753–753. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 37638: sources/stacks-more-algebra/pinned-original-source.tex, source lines 37638–37638. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 37675: sources/stacks-more-algebra/pinned-original-source.tex, source lines 37675–37675. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 37847: sources/stacks-more-algebra/pinned-original-source.tex, source lines 37847–37847. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let R be a PID. Every finite R-module M is of isomorphic to a module of the form*

$$R^{\oplus r} \oplus \bigoplus\nolimits_{i=1,\ldots,n} R/f_iR$$

for some $r, n \geq 0$ and $f_1, \ldots, f_n \in R$ nonzero.

Proof. A PID is a Noetherian Bézout ring. By Lemma 37847 it suffices to prove the result if M is torsion. Since M is finite, this means that the annihilator of M is nonzero. Say $fM = 0$ for some $f \in R$ nonzero. Then we can think of M as a module over R/fR. Since R/fR is Noetherian of dimension 0 (small detail omitted) we see that $R/fR = \prod R_j$ is a finite product of Artinian local rings R_i (Algebra, Proposition 00KJ). Each R_i, being a local ring and a quotient of a PID, is a generalized valuation ring in the sense of Lemma 37638 (small detail omitted). Write $M = \prod M_j$ with $M_j = e_jM$ where $e_j \in R/fR$ is the idempotent corresponding to the factor R_j. By Lemma 37675 we see that $M_j = \bigoplus_{i=1,\ldots,n_j} R_j/\overline{f}_{ji}R_j$ for some $\overline{f}_{ji} \in R_j$. Choose lifts $f_{ji} \in R$ and choose $g_{ji} \in R$ with $(g_{ji}) = (f_j, f_{ji})$. Then we conclude that

$$M \cong \bigoplus R/g_{ji}R$$

as an R-module which finishes the proof. □